

Materials & Methods

1 Software used

The model was implemented in NetLogo version 7.0.3 using the array extension (<https://www.netlogo.org/>). Parameter sweeps were executed using NetLogo’s BehaviorSpace tool. Output data were exported as CSV files and analysed in Jupyter Notebook using Python 3.12.7 with the packages pandas, NumPy, SciPy, statsmodels, Seaborn and Matplotlib. Simulations and analyses were executed on a MacBook Air with an Apple M1 processor and 8 GB RAM.

2 Model overview

The model consists of a population of 200 cooperative agents belonging to fixed social groups ($n = 2$). Each agent possesses a unique multi-feature linguistic repertoire, derived from a small state space. Agents interact pairwise, and accumulate fitness through their cooperation with other agents.

While cooperation with in-group is rewarded, agents are punished if they cooperate with the out-group; likewise, out-group defection is rewarded while punished with in-group members. Group identity is not usually observable in interactions, and when paired with a stranger agents must use either a prior-based heuristic or Bayesian inference to decide whether or not to cooperate.

Agents adapt to their environment through learning. Partial observation of true group identities of stranger allows agents to map out the interaction structure and learn variant-group associations, while they update their language via social learning from higher-fitness group members.

Variable	Description
<code>group-id</code>	Social group membership
<code>language</code>	Linguistic repertoire
<code>fitness</code>	Accumulated fitness from interactions
<code>knowns</code>	Familiar in-group interaction partners
<code>stranger-counts</code>	Interaction experience counts
<code>ingroup-stranger-counts</code>	
<code>p-stranger-est</code>	Interaction structure priors based on counts
<code>p-ingroup-est</code>	
<code>language-counts</code>	Variant-group association counts
<code>risk-perception</code>	Estimated interaction risk (based on priors)

Table 1: Agent-level variables

3 Agent representation

Table 1 shows the agent-level variables.

4 Initialisation

At setup, the following initialisation steps are performed:

1. A population of `pop-size` agents is created.
2. Agents are divided equally between groups.
3. Agents are assigned a random linguistic repertoire (`language`).
4. Agents initialise their interaction prior counts and Bayesian language counts (array) to the values of the Dirichlet priors.
5. Agents add a random in-group member to their `knowns` agenset.

5 Interaction dynamics

5.1 Partner selection

Interaction partners are chosen probabilistically. This is implemented in NetLogo by generating a random number: if lower than `p-outgroup`, the out-group contact rate, a random agent from this group is chosen; else, a random in-group member is chosen, which may or may not be familiar to the agent.

5.2 Strategy choice

Partners in an agent’s **knowns** are cooperated with automatically. With stranger partners, the decision is divided into two stages: strategy selection, and cooperation action. Agents choose between heuristic and inference strategies probabilistically according to a logistic function. The expected utility of each action is calculated as described in the Model Design section. Let ΔEU_{strat} denote the difference between the expected utility of inference and the expected utility of the heuristic strategy. The probability of selecting inference is given by

$$P(\text{inference}) = \frac{1}{1 + e^{-s\Delta EU_{strat}}}$$

where s (**decision-steepness**) controls the sensitivity of agents to expected utility differences. Our model used a steep curve ($s = 15$), yielding a sharp transition between strategies.

For each group and linguistic feature, agents maintain Dirichlet counts representing variant-group associations. These counts are converted to conditional probabilities $P(v_f|g)$ when inference is performed. To ensure numerical stability, Bayesian inference is implemented in log space. The product over features is replaced by a sum of log-likelihoods:

$$\log P(g | s) \propto \log P(g) + \sum_f \log P(v_f | g)$$

where s denotes the observed linguistic signal and v_f the variant observed for feature f . These log-scores are converted into probabilities via a softmax transformation over groups:

$$P(g | s) = \frac{e^{\text{score}(g)}}{\sum_{g'} e^{\text{score}(g')}}$$

This is equivalent mathematically to direct Bayesian normalisation but avoids numerical underflow from multiplying small probabilities.

5.3 Cooperation decision

The final cooperation decision is calculated probabilistically using a logistic function, as laid out in the Model Design. The expected utility of cooperation is calculated using the equation shown in equations (2–3) in that section, using the in-group stranger prior alone for the heuristic strategy, or the posterior probability based on observed signals while

using Bayesian inference. The differential between the expected utility of cooperating and of defecting (ΔEU_{dec}):

$$P(\text{cooperate}) = \frac{1}{1 + e^{-s\Delta EU_{dec}}}$$

where s is controlled by **decision-steepness**, as before.

5.4 Interaction outcome

Agents' **fitness** is increased by **payoff** when they engage in adaptive cooperating behaviour (in-group cooperation, out-group defection). Maladaptive behaviour (in-group defection, out-group cooperation) incurs a loss to fitness (**penalty**). When interacting with in-group partners, agents receive a secondary payoff or penalty when their partners cooperate or defect, respectively.

5.5 Diffusion

There is a low probability (**diffusion-rate**) per interaction that agents copy a random variant from their interaction partner.

6 Learning

6.1 Interaction structure

Agents store counts based on prior interaction experience, forming a beta distribution. Agents store counts based on two types of interaction partners: strangers (both groups), and specifically in-group strangers. Counts are stored in pairs: α (occurrence of given type) and β (not the given type).

Stranger counts are incremented by one at each timestep; in-group stranger counts are only updated when true group identity is revealed (at a rate of **p-observe**). These counts are then converted into probabilities, the priors **p-stranger-est** and **p-ingroup-est**, using the mean (μ) formula:

$$P(\alpha) = \frac{\alpha}{(\alpha + \beta)}$$

An agent’s risk perception is calculated as described in the Model Design using these priors.

6.2 Variant-group learning

Given they rely on observing true group identity, variant-group counts are only updated at a rate of `p-observe`. Each feature-variant-group position observed is located in the agent’s array and incremented by 1. This distribution is used to calculate the probabilities for Bayesian inference.

7 Social learning

Social learning takes place within known networks. A random percentage in the range `[0, social-learning-rate]` of the worst performing agents by fitness is selected to learn at each timestep. A higher-fitness agent is chosen from their known network to serve as a model. The learners adopt the models’ full `language`, with one feature being set to a random variant a a likelihood of `noise-rate`.

8 Fitness decay

Fitness is decayed at each timestep, ensuring it does not spiral endlessly. After the social learning phase, all agents’ fitness is set to 95% of its previous level.

9 Simulation schedule

The full simulation schedule is as follows:

1. Interactions
2. Belief updates
3. Social learning
4. Mutation
5. Fitness decay

10 Experimental design

10.1 Parameter sweeps

All parameter sweeps were executed using NetLogo’s BehaviorSpace tool. The experimental parameters were varied as follows:

- `p-outgroup`: [0.01, 0.1, 0.2, 0.3, 0.4, 0.5]
- `social-memory-size`: [10, 25, 50, 75, 90, 100]

Each parameter combination was simulated 30 times for 3000 timesteps, which was more than enough time for the model dynamics to stabilise. Simulation outputs were exported to CSV files. Statistical analyses were performed on parameter-level aggregates computed over the final 500 timesteps of each run.

10.2 Other experiments

The model variants used to isolate inference’s role in the causal chain, described in the Causal Intervention section, used identical parameter values as the main model, varying only the availability of cooperating strategies to agents. This bypassed the strategy choice, with agents always using either the heuristic or inferential strategies, depending on the variant. In this case, the only metric recorded was global Jensen-Shannon divergence.

The time series data were obtained from two parameter regimes, corresponding to a high-risk regime (`p-outgroup` = 0.3, `social-memory-size` = 20) and a low-risk regime (`p-outgroup` = 0.05, `social-memory-size` = 40). These models also ran for 3000 timesteps, recording metrics at each tick.

10.3 Output metrics

Linguistic differentiation between groups was quantified using the Jensen–Shannon (JS) divergence between group-level linguistic distributions. For each group, frequencies were calculated separately for every feature–variant combination and pooled into a single probability distribution of length $(F \times V)$, where (F) is the number of linguistic features and

(V) the number of variants per feature. Given two group distributions (Q) and (R), the midpoint distribution was defined as

$$M = \frac{Q + R}{2}$$

JS divergence was then calculated as

$$JS(Q, R) = \frac{1}{2}D_{KL}(Q \parallel M) + \frac{1}{2}D_{KL}(R \parallel M)$$

where D_{KL} denotes the Kullback–Leibler divergence. Values were normalised by $\ln 2$, producing a measure bounded between 0 and 1, where 0 indicates identical linguistic distributions and 1 maximal divergence.

Feature-level JS divergence was calculated using the same procedure, except that distributions were constructed from the variants of a single feature rather than all feature–variant combinations. For a feature with (V) variants, the resulting distributions therefore had length (V). The minimum, median and maximum feature-level JS divergence across all features were recorded.

Other metrics recorded included mean **risk-perception** and the proportion of inference used in stranger interactions.

11 Parameter table

Parameter	Description	Value
<i>Structural parameters:</i>		
pop-size	Population size	200
num-groups	Number of social groups	2
num-features	Number of linguistic features per agent	3
num-variants	Number of linguistic variants per feature	3
interaction-rounds	Number of interactions per timestep	3
social-memory-size	Maximum number of individuals an agent can be familiar with	sweep ^a
<i>Interaction parameters:</i>		
p-outgroup	Out-group contact rate	sweep ^a
inference-cost	Cognitive cost of inference in expected utility calculation	0.3
decision-steepness	Logistic sensitivity in strategy choice	15
payoff	Reward for adaptive cooperation	1
penalty	Penalty for maladaptive cooperation	1
<i>Learning parameters:</i>		
p-observe	Probability of observing true group identity	0.2
social-learning-rate	Max proportion of agents updated per timestep	0.05
noise-rate	Mutation probability during social learning	0.33
diffusion-rate	Probability of feature copying during interaction	0.001
Initial pseudocounts		1
Learning increment (counts)		1
<i>Experimental design:</i>		
Simulation length		3000 ^b
Analysis window (final)		500 ^b
Runs per condition		20

^a Parameter experimentally varied across a range of values.

^b Timestep (tick) values